

COMMON CONTROL SYSTEM (CCS)

Site Installation and Configuration Guide

Software Release 4.2.0 (illustrative)

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Approval

This guide is approved for release with CCS software release 4.2.0 (illustrative).

Role	Name and title	Signature	Date
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Record of Changes

Each revision adds a row. Change bars in the page margins mark text changed since the prior revision.

Rev.	Date	Description of change	Pages affected	Entered by
A	23 Sep 2026	Initial release for CCS 4.2.0 (illustrative)	All	NBS

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Preface

Purpose. This guide tells site administrators how to install, configure, verify, and record CCS software release 4.2.0 on a control-station host.

Scope. New installations and upgrades on land, ship, and submarine control stations. Operator use of the software is covered in the CCS Software User Manual.

Audience. Site administrators with root access to the station host. The Information System Security Officer (ISSO) witnesses the media check and signs the installation record.

How this guide is organized. Sections 1 and 2 prepare you. Sections 3 through 7 are the installation, in the order you perform it. Section 8 covers upgrades and rollback. Appendix A is troubleshooting; Appendix B defines terms.

Conventions

Table P-1. Typographic and safety conventions

Convention	Meaning
Bold	A control on screen, such as Validate
Monospace	A command, file name, or value you type
<i>Italic line after a step</i>	What you should see when the step works
WARNING	Risk of injury, loss of a vehicle, or mission failure
CAUTION	Risk of damage to equipment or loss of data
NOTE	Information that helps you do the task

Related documents

Table P-2. Related documents and their purpose

Document	Purpose
CCS Version Description Document (VDD), release 4.2.0	Lists the exact files, versions, and checksums on the release media
CCS Software User Manual, Sections 3.3 and 4.1.3	Software environment; operator view of installation
Site configuration baseline	The approved site property values
DoD Instruction 5200.48, Controlled Unclassified Information	Marking of media, logs, and printed records

1 Overview

This guide takes a site administrator from verified release media to a working, tested CCS station in about 3 hours. Perform the five stages in Figure 1-1 in order; each ends with a check that must pass before the next begins.

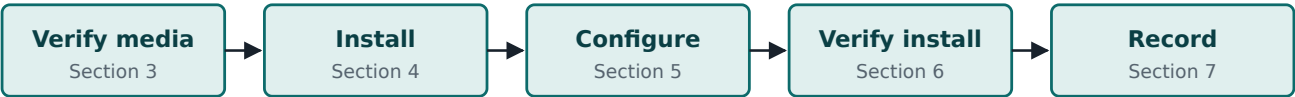


Figure 1-1. Installation stages

1.1 Roles and time

Table 1-1. Roles, time, and result

Item	Detail
Performs	Site administrator
Witnesses and signs	ISSO
Time required	About 3 hours: 30 min media check, 45 min install, 60 min configure, 45 min verify
Result	CCS installed, configured, passing health checks, and recorded in the site CM log

2 Before You Begin

Confirm every requirement in Table 2-1 before you open the media. A missing prerequisite is the most common cause of a failed installation.

Table 2-1. Prerequisites

Requirement	Minimum (illustrative)	How to check
Operating system	RHEL 9.x, STIG-hardened site image	<code>cat /etc/redhat-release</code>
Java runtime	OpenJDK 17	<code>java -version</code>
Memory / disk	32 GB RAM; 200 GB free on /opt	<code>free -g; df -h /opt</code>
Displays	Two monitors, 1920 × 1080 or higher	Station display settings
Accounts	Admin account with sudo; ccs service account	<code>id ccs</code>
Media	Release 4.2.0 disc, labeled and logged	Label matches the VDD
Site baseline	Approved site.properties values	Signed copy from site CM

WARNING

Install and verify with the datalink disconnected or in simulation mode. A station that is half-configured can send invalid commands to a live vehicle.

3 Verify the Release Media

Prove the media is exactly what the VDD describes before anything touches the station.

1. Check the disc label against the VDD: release **4.2.0**, build number, date, and classification marking must all match.
2. Mount the disc read-only:

```
sudo mount -o ro /dev/sr0 /mnt/ccs-media
```

3. **CAUTION**
If any checksum fails, stop. Do not install from that media.

Verify the checksums against the list on the media:

```
cd /mnt/ccs-media  
sha256sum -c SHA256SUMS
```

Every line ends in OK.

4. Compare the checksum of the `SHA256SUMS` file itself with the value printed in the VDD.
Values match exactly.
5. Record the media control number in the site media log. The ISSO initials the entry.

4 Install the Software

1. Back up the current installation, if one exists:

```
sudo /opt/ccs/bin/ccs-backup --out /var/backups/ccs-$(date +%F).tar.gz
```

2. Run the installer from the media:

```
cd /mnt/ccs-media  
sudo ./install.sh --release 4.2.0 --prefix /opt/ccs
```

The installer prints `Install complete: CCS 4.2.0-1187.`

3. Confirm the service is registered but not yet running:

```
systemctl status ccs
```

Status shows `loaded, inactive (dead)`. *The service starts only after configuration.*

4. Unmount and remove the media; return it to storage and log the return.

5 Configure the Site

All site-specific settings live in one file, `/opt/ccs/conf/site.properties`. Copy values from the approved site baseline; do not invent them. Table 5-1 explains each property.

Table 5-1. Site properties

Property	What it actually is	Example value
<code>ccs.site.id</code>	The station's unique name on the network	<code>SITE-SD-02</code>
<code>ccs.station.type</code>	Where the station lives; sets UI layout and defaults	<code>land ship sub</code>
<code>ccs.platforms.enabled</code>	Which vehicle types this station may control	<code>mq4c,uuv</code>
<code>ccs.datalink.mode</code>	Simulated or live vehicle connection	<code>sim</code> until verified
<code>ccs.msg.gateway</code>	Address of the external messaging gateway	<code>msg-gw.site.local:7443</code>
<code>ccs.log.dir</code>	Where logs are written; must be on marked storage	<code>/var/log/ccs</code>
<code>ccs.ui.banner</code>	Classification banner text shown on every screen	Per site baseline

1. **CAUTION**

Leave `ccs.datalink.mode=sim` until Section 6 is complete. Changing it early connects the station to the live datalink.

Open the file with the site-approved editor:

```
sudo -u ccs vi /opt/ccs/conf/site.properties
```

2. Set each property to the baseline value. A completed file looks like this:

```
# site.properties -- CCS 4.2.0 -- SITE-SD-02 (illustrative)
ccs.site.id=SITE-SD-02
ccs.station.type=land
ccs.platforms.enabled=mq4c,uuv
ccs.datalink.mode=sim
ccs.msg.gateway=msg-gw.site.local:7443
ccs.log.dir=/var/log/ccs
ccs.ui.banner=<per site baseline>
```

3. Validate the file before starting the service:

```
sudo /opt/ccs/bin/ccs-config --check
```

Output ends with Configuration valid (7 of 7 required properties set). Correct any property it names, then check again.

6 Verify the Installation

1. Start the service:

```
sudo systemctl enable --now ccs
```

2. Run the health check:

```
sudo /opt/ccs/bin/ccs-health
```

All components report UP: core, planning, vehicle-mgmt, mission-mgmt, messaging.

3. Log in at the station with a test operator account.

The CCS window opens with the correct classification banner and site ID.

4. Run the simulation smoke test: open `SMOKE-01.plan`, select **Validate**, then **Launch (Sim)**.

The simulated vehicle follows the route and reports status for 5 minutes with no errors.

5. **WARNING**

Do not change the datalink mode until steps 1 through 4 pass and the ISSO approves.

Set `ccs.datalink.mode=live` and restart the service.

7 Record the Installation

An installation is not complete until it is in the CM record. Complete Table 7-1 and file it with the site configuration log.

Table 7-1. Installation record

Field	Entry
Site ID	
Release and build installed	
Media control number	
Checksum verification (pass / fail)	
Configuration check result	
Health check result	
Smoke test result	
Datalink mode at completion	
Installed by (name, date)	
ISSO (name, date)	

8 Upgrade and Rollback

Upgrades repeat Sections 3 through 7 in full. If the health check or smoke test fails and cannot be corrected within the maintenance window, roll back:

1. Stop the service: `sudo systemctl stop ccs`
2. Restore the backup made in Section 4, step 1:

```
sudo /opt/ccs/bin/ccs-restore --in /var/backups/ccs-<date>.tar.gz
```

3. Start the service and repeat the health check. Record the rollback and its reason in the CM log.

Appendix A — Troubleshooting

Find the symptom in Table A-1. If none matches, contact site CM.

Table A-1. Troubleshooting

Symptom	Likely cause	Remedy
Checksum reports FAILED	Damaged or wrong media	Stop. Do not install. Report to site CM for replacement media.
Installer: Java 17 not found	Wrong or missing runtime	Install the site-approved OpenJDK 17 package; rerun the installer.
<code>ccs-config --check</code> names a property	Missing or misspelled value	Correct it from the site baseline; check again.
Health check: messaging DOWN	Gateway address or firewall	Confirm <code>ccs.msg.gateway</code> ; ask the network administrator to confirm port 7443 is open.
No classification banner on screen	<code>ccs.ui.banner</code> is empty	Set it from the baseline; restart the service.
Smoke-test vehicle does not move	Plan not validated, or datalink not in <code>sim</code>	Validate the plan; confirm <code>ccs.datalink.mode=sim</code> .

Appendix B — Glossary and Acronyms

Table B-1. Glossary and acronyms

Acronym	Term	What it actually is
CCS	Common Control System	One common software suite that plans, controls, and manages multiple Navy unmanned vehicles
CM	Configuration Management	Keeping a controlled, recorded baseline of exactly which versions are installed where
CUI	Controlled Unclassified Information	Unclassified information that still requires safeguarding and marking under DoDI 5200.48
ISSO	Information System Security Officer	The person responsible for a system's security posture; witnesses media checks and signs records
STIG	Security Technical Implementation Guide	DoD configuration standard used to harden the host operating system
VDD	Version Description Document	The document listing what is in a release: files, versions, checksums, known issues

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